

Q1 (2021 T1 Final Exam Q4a)

A regular household system of a single-phase three-wire circuit allows the operation of both 120 V and 240 V, 60 Hz appliances. The household circuit is shown in Figure 1. Calculate the current through each of the appliances (i_{lamp} , i_{ref} and i_{range})

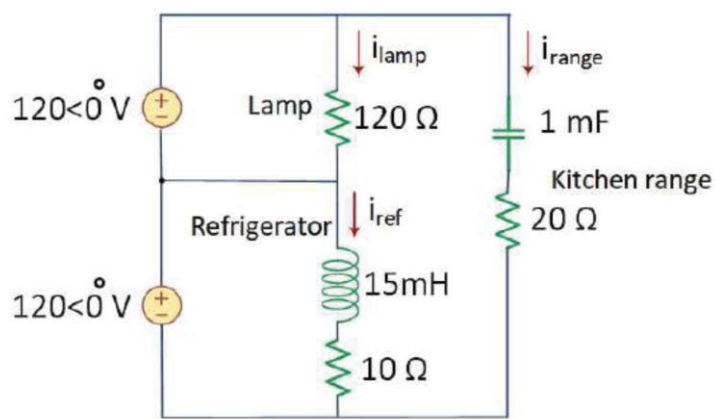


Figure 1

Answers:

$$i_{lamp} = \cos(120\pi t) \text{ A}, i_{ref} = 10.45\cos(120\pi t - 29.47^\circ) \text{ A}, i_{range} = 11.9\cos(120\pi t + 7.55^\circ) \text{ A}.$$

Q2 (Modified from 2023 T1 Final Exam Q4)

The circuit in Figure 2 is a high pass filter that is connected to a tweeter.

i) Calculate the output voltage $v_{out}(t)$.

ii) Calculate the average power absorbed by the tweeter (i.e., 4Ω resistor).

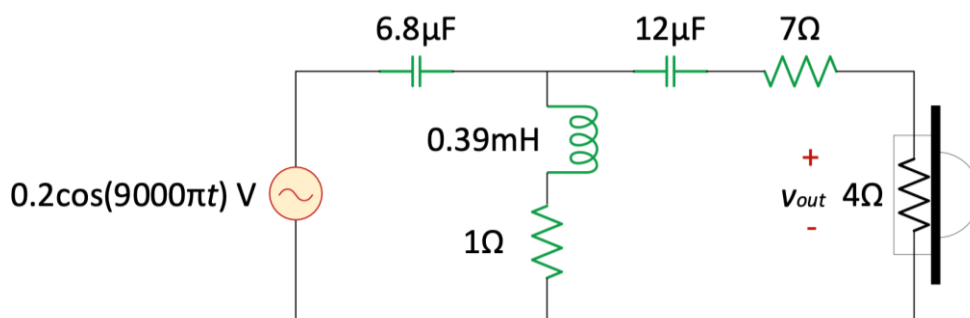


Figure 2

Answers: i) $v_{out}(t) = 0.0867 \cos(9000\pi t + 51.6^\circ) \text{ V}$, ii) 0.94 mW

Q3 (2022 T1 Final Exam Q4a)

The circuit shown in Figure 3 is used in a radio receiver to filter a specific interfering signal. If the amplitude of the input signal is 1V,

- Derive the expression of the output voltage phasor V_{out} as a function of the angular frequency.
- Calculate the angular frequency for which the equivalent impedance of the filter between terminals a-b, Z_{eq} , is purely resistive. Note that this angular frequency is the frequency of the interfering signal.
- Substitute the value of angular frequency obtained in part ii) in your derived output voltage phasor V_{out} from part i) to find the value of V_{out} . Based on this result, explain why the circuit shown in Figure 3 is effectively removing the interfering signal.

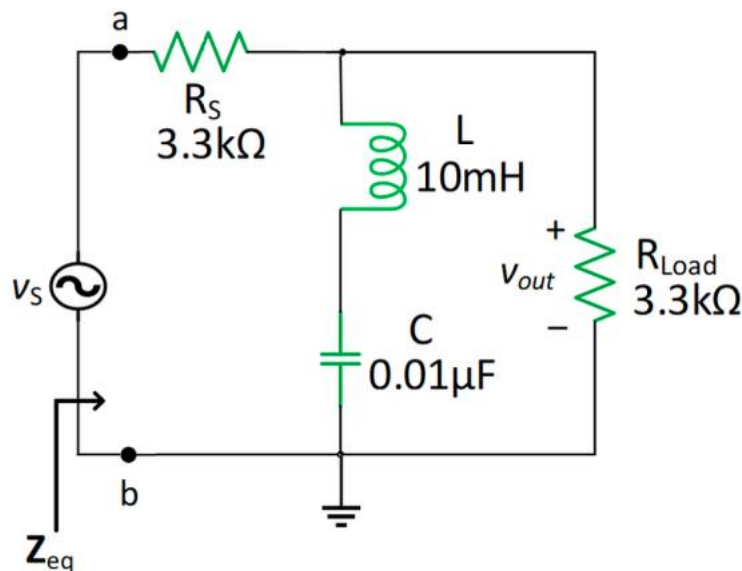


Figure 3

Answers:

$$i) v_{out} = \frac{j(33\omega - \frac{33 \times 10^{10}}{\omega})}{3300^2 + j2(33\omega - \frac{33 \times 10^{10}}{\omega})}$$

$$ii) \omega = 10^5 \text{ rad/s}$$

$$iii) v_{out} = 0 \text{ V}$$

If the frequency of the interfering signal is equal (or close) to the one calculated in ii), the amplitude of the output becomes 0 (or close), which means that the interference is filtered/removed (or attenuated).

Q4 (2021 T1 Final Exam Q4b)

Calculate $v_o(t)$ in the circuit of Figure 4.

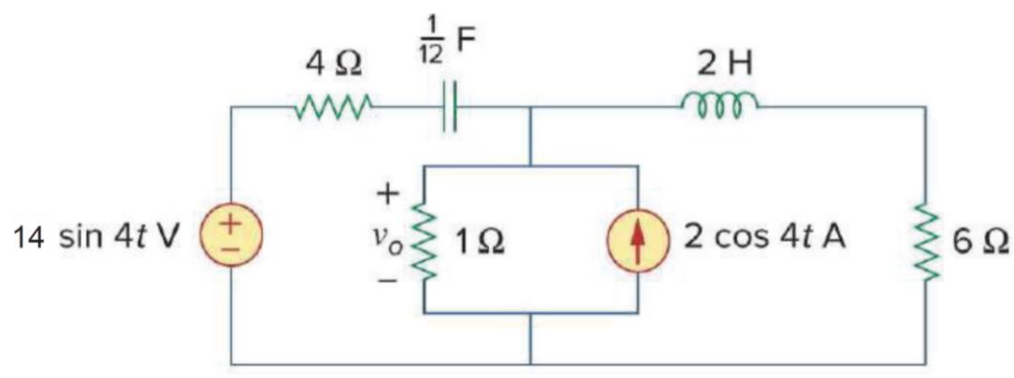


Figure 4

Answer: $v_o(t) = 3.53 \cos(4t - 33.21^\circ) \text{ V}$